

Tau3 Banking - GLM-5.2 Trial-0 Parity Report

Agent-owned audit of the completed 97-task single-trial run against Sierra's official four-trial GLM-5.2 submission.

144/388 OFFICIAL PASSES	33/97 REPRODUCED PASSES	-3.0928 pp RATE DELTA	97/97 VALID COVERAGE
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NOT EXACT

TL;DR

Official	Reproduced	Delta	Coverage	Main attention
144/388 , 37.1134%	33/97 , 34.0206%	-3.0928 percentage points	97/97, all <code>user_stop</code> , 0 infrastructure errors	Three official trials are missing, and official-vs-candidate traces have not been audited end to end

The run is execution-complete and its 33/97 score is valid for the frozen single-trial contract. It is **not** an exact reproduction of the official 144/388 result: the denominator and rate differ, while route, sandbox, embedding-cache, tool-usage, and trace-rendering parity remain unresolved.

Parity evidence

Layer	Status	Evidence and nuance
Score	FAIL	Official 144/388 = 37.11340206185567%; reproduced 33/97 = 34.02061855670103%. Rate delta: -3.09278350515464 percentage points.
Harness	PARTIAL	Both use tau2-bench v1.0.1, 97 tasks, <code>alltools</code> , seed 300, 200 steps, and 10 consecutive errors. Official has four trials; reproduced has one and uses Modal, whose exact command-output equivalence is unproven.
Tools	PARTIAL	Reproduction exposes the frozen 17-tool initial schema, hash <code>5e2d300e...271ae1</code> . GLM trace-level used-tool counts, arguments, outputs, and discoverable-tool order have not been compared.
Model config	PARTIAL	Reproduction requested <code>openrouter/z-ai/glm-5.2</code> , temperature 1.0, top-p 1.0, 16,384 output tokens, and <code>xhigh</code> reasoning; all 2,450 assistant responses resolved through Sail Research. Official provider-route identity still needs trace-level verification.
Trace rendering	PARTIAL	<code>tau2.to_litellm_messages</code> , OpenAI function-tool JSON, and <code>tool_choice=auto</code> ; reasoning is raw-trace-only after its turn. GLM tokenizer, Jinja template, BOS/EOS policy, and hosted renderer are not exposed or hashed.
Coverage	PASS	97/97 exact task keys, 97 <code>user_stop</code> terminations, zero final infrastructure placeholders, unique response-ID receipts, and clean-runtime validation.

Layer	Status	Evidence and nuance
Grading	PARTIAL	Deterministic grading and reward recombination passed internal guards. Task 102 used the dated GPT-4.1 judge and its NL assertion passed, while DB failure kept reward 0. The complete official per-task component vector has not yet been compared.
Behavior	NOT COMPARED	Official trajectories are public, but a strict GLM official-vs-candidate message, tool, output, and reward-vector audit has not yet been published.

Trace boundary finding: the checked-in renderer serializes assistant content and tool calls into the next request, but not prior `reasoning_content`; the complete provider response remains in `raw_data`. This proves the reproduction's history policy. It does not prove that Sierra's hosted endpoint applied the same tokenizer revision or private Jinja template, so that layer remains partial.

Result and execution metrics

Metric	Reproduced value	Scope
Valid trajectories	97/97	One trial
Passed trajectories	33	Binary task reward
Assistant responses	2,450	GLM-5.2 through Sail Research
User-simulator responses	708	GPT-5.2 through OpenAI/default
Serialized cost breakdown	\$21.61216452 + \$1.93359775 + \$0.11268	Assistant + user simulator + judge
Serialized total	\$23.65844227	Excludes embeddings, Modal, and discarded attempts

Resume retried infrastructure keys only; it never replayed or transplanted scored outputs. Thirty-six empty terminal responses and nine transient TLS failures were discarded, so their spend is absent from `results.json`.

Exact commands

Run from `/home/tianhaowu/mono/tau3-banking` on the evaluation host. The OpenRouter credential is read from `~/rllm/config.json`; it must never appear in argv, logs, manifests, or this report.

1. Validate the route with the bound smoke

```
uv run --frozen --extra knowledge python \
  reproduction/tau3_banking/run_glm52.py smoke \
  --output-dir \
  reproduction/tau3_banking/runs/glm52_smoke_8904806_20260823T0415Z \
  --execute --confirm-paid-api-calls
```

The smoke must complete task 001, resolve GLM-5.2 only through Sail Research, resolve GPT-5.2 only through OpenAI, and pass route, grading, response-ID, provenance, and cost guards before a full run is accepted.

2. Launch or resume the frozen 97-task run

```
uv run --frozen --extra knowledge python \
  reproduction/tau3_banking/run_glm52.py full \
  --output-dir \
  reproduction/tau3_banking/runs/glm52_full_8904806_20260823T0420Z \
  --smoke-manifest \
  reproduction/tau3_banking/runs/glm52_smoke_8904806_20260823T0415Z/glm52_manifest.json \
  --resume --execute --confirm-paid-api-calls
```

- Remove `--resume` only for the first launch into a new empty run directory.
- `--resume` authenticates the immutable checkpoint, skips validated keys, and retries only infrastructure or missing keys.
- `--confirm-paid-api-calls` acknowledges paid work; the credit gate is admission control, not a provider hard cap.
- Do not change the task list, smoke manifest, runtime commit, model routes, or sampling configuration within a run directory.

3. Render and verify this PDF

```
uv run reproduction/tau3_banking/render_parity_report.py \
  reproduction/tau3_banking/reports/glm52_trial0_parity.md \
  reproduction/tau3_banking/reports/glm52_trial0_parity.pdf \
  --pages-dir \
  reproduction/tau3_banking/reports/glm52_trial0_parity-pages
```

This command checks unresolved placeholders, external URI annotations, extractable text, replacement glyphs, page count, and nonempty page images. Every rendered page must still be inspected visually.

Links and attention

- [Official immutable submission](#), [raw official trajectories](#), and [trajectory visualizer](#)
- [Benchmark at the pinned upstream commit](#), [official tasks](#), [tool implementation](#), and [grader](#)
- [GLM-5.2 endpoint](#), [runner](#), [result receipt](#), [Modal receipt](#), and [reproduction log](#)
- [PDF source](#) and [PDF QA receipt](#). Artificial Analysis's separate [27% result](#) uses a non-comparable x5 `bm25_grep`/GPT-5.4 Mini contract.

Attention: three more trials are necessary but not sufficient. Exact parity also requires 144/388, matching task/trial rewards, routes, trace/tool behavior, renderer evidence, sandbox, and embedding state.

Hashes: result `84c89c9a0a87a5aaf0c6a16998e4333a860ae3d810c815690cf6619c96667774`; manifest `639cc89544a4aa72bf7c16c89770825cb406ed00f6f7e16e7effbaa6ee0ea942`. Remaining receipts are in the linked JSON; the 329 MB official artifact was linked, not copied.

Evaluation runtime commit `4643dc52d58538e5e897b761d6e061f2edc88a49`; final validation completed 2026-08-23 UTC.